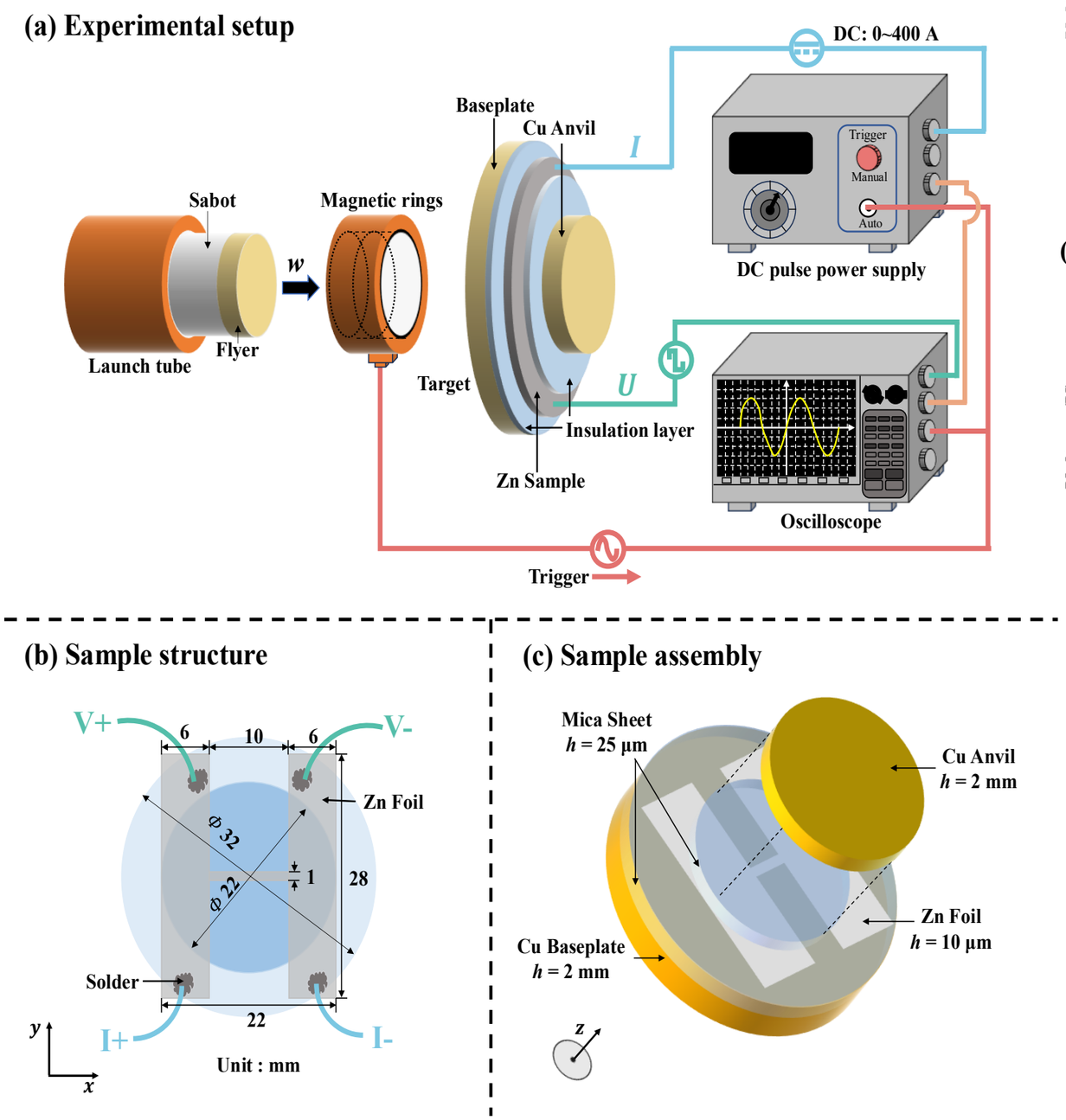
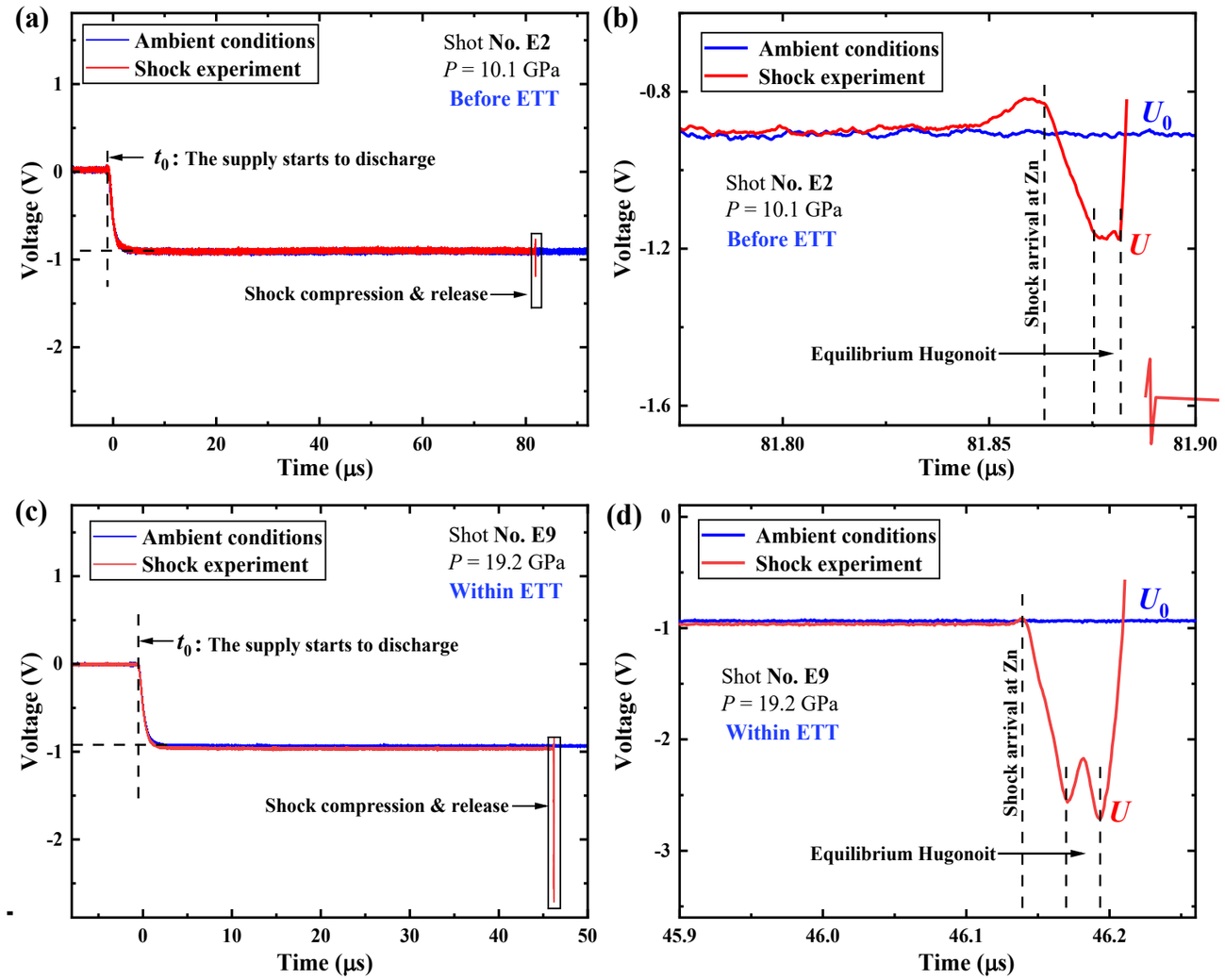
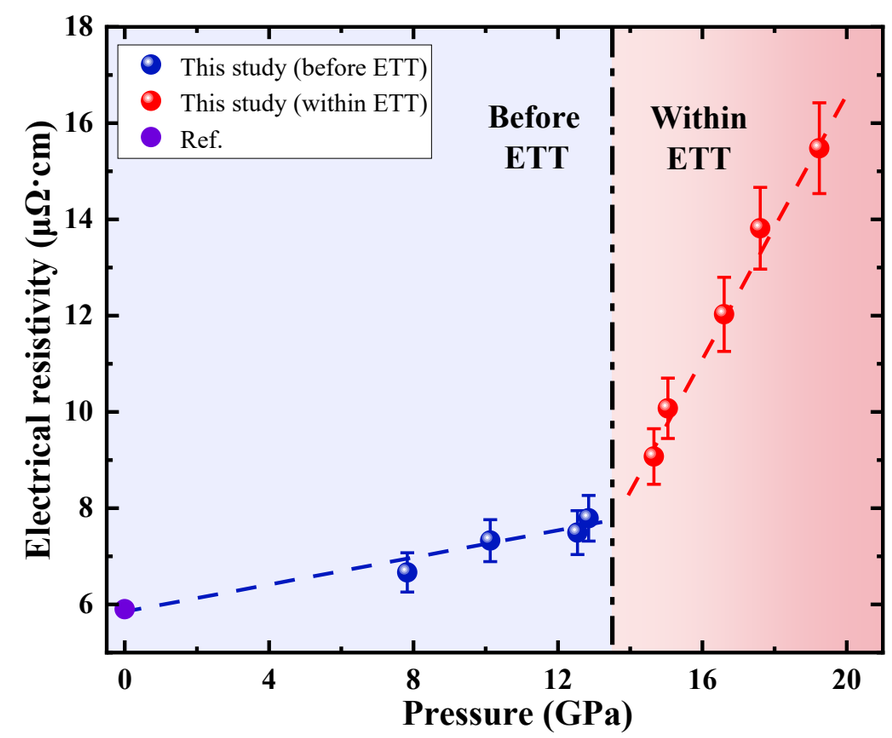

(a) Experimental setup
DC: 0~400 A
Baseplate
Cu Anvil
I
Sabot
Magnetic rings
w
Flyer
Launch tube
Target
U
Insulation layer
Zn Sample
DC pulse power supply
Trigger
Manual
Auto
Oscilloscope
Trigger
(b) Sample structure
V+
V-
6
10
6
Zn Foil
Φ 32
Φ 22
1
28
Solder
22
I+
I-
Unit : mm
y
x
(c) Sample assembly
Mica Sheet
h = 25 μm
Cu Anvil
h = 2 mm
Zn Foil
h = 10 μm
Cu Baseplate
h = 2 mm
z


(a)
Ambient conditions
Shock experiment
Shot No. E2
P = 10.1 GPa
Before ETT
t0 : The supply starts to discharge
Shock compression & release
Voltage (V)
Time (μs)
(b)
Ambient conditions
Shock experiment
U0
U
Shot No. E2
P = 10.1 GPa
Before ETT
Shock arrival at Zn
Equilibrium Hugonoit
Voltage (V)
Time (μs)
(c)
Ambient conditions
Shock experiment
Shot No. E9
P = 19.2 GPa
Within ETT
t0 : The supply starts to discharge
Shock compression & release
Voltage (V)
Time (μs)
(d)
Ambient conditions
Shock experiment
U0
U
Shot No. E9
P = 19.2 GPa
Within ETT
Shock arrival at Zn
Equilibrium Hugonoit
Voltage (V)
Time (μs)


This study (before ETT)
This study (within ETT)
Ref.
Before ETT
Within ETT
Electrical resistivity (μΩ·cm)
Pressure (GPa)